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Anthropogenic activity causing microbial changes in the water of Kanke dam. Ranchi, Jharkhand

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Abstract : The present work is an attempt to analyze various microbial factor and the a change occure in phytoplankton,zooplankton and microphytes due to pollution and calculated by the method of Welch[1]. Kanke dam is one of the important water reservoir of ranchi through which drinking water is supplied to various parts of ranchi city.Recently, due to lack of proper attention & maintenance and anthropogenic activities the water of reservoir is getting polluted & causing health problems such as dysentery, diarrhea, jaundice & other water borne diseases. The source of pollution are flow of drains, it resulted in deposition of undersirable inorganic & organic components.

Keywords: Biotic, abiotic, Bod, phytoplanktons, anthropogenic, zooplankton, macrophytes.

INTRODUCTION

Pollution of water is a global problem. Any undesirable change in properties of water which affects the quality of water and makes it unfit for the useful purpose is called pollution. Biological water cycle involves the exchange of water between abiotic and biotic factors of the environment. In the terrestrial ecosystem, plants absorb water from soil and animals take water directly, along with food. Some part of this water is retained by the living cells and the rest is released. Besides these physical, chemical and biological properties water has played a very significant role in human settlement and civilization. That is why ,most of the ancient civilization like, Indus,Egyptian,Mesopotamian civilization etc. had developed on the banks of rivers.

ANTHROPOGENIC POLLUTION. Bio pollutants are sewages, municipal wastes, factory discharges, and aquatic micro-organisms[2]. Some of the wastes which are discharged into the air or land may also eventually reach the wastes and discharges are making the problem grimmer. Not only the fresh water ,but also marine water is being polluted by the disposal of large amount of sewage ,garbage, radioactive wastes and industrial effluents into the ocean.

EFFECT OF POLLUTON. Aquatic ecosystems have self regulating mechanism to over come this problem if ,the level of pollutants in water is within the limit . When this level increases above the threshold limit, water becomes rich in nutrients, and makes water body highly productive and eutrophic. Very thick mat of algal blooms may be observed on the surface of such water bodies .Some microorganisms, which were converting these wastes into minerals now, they are unable to do so.

BIOTIC COMPONENTS. It includes producer, consumers and decomposers. Producers are mainly autotrophic organisms, consumers and decomposers. A biotic component mainly include organic and inorganic. According to Wetzel, limnology in broad sense is the study of the functional relationship and the productivity of fresh water biotic communities ,which are affected by the dynamic of physical ,chemical and biotic environmental parameters[3]. India is a tropical county, soils freshwater resources show tremendous ecological diversity in temperature, salt types, content concentration, turbidity.

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These are influenced by all physical chemical and biological factorslike, temperature[4], turbidity, alkalinity[5], density, carbonates, bicarbonates, nitrates, phosphates, silicates, chlorides, transparency, conduvtivity, Co_2, dissolve Oxygen[6] pH[7], BoD[8], COD, and Micronutrients[9]. Besides these, micro and macro organism also influence their ecosystem.

NEED OF STUDY. The present work 'Studies on the Biotic[10] and Abiotic factors[11] causing pollution of water , Kanke Dam is an attempt to analyse Biotic & Abiotic factors of the Dam , as it is widely used by a large number of people and has not yet been explored scientifically. Huge populations of Ranchi city use the Dam widely for drinking ,bathing, wasing, fishing etc Rain water is the only source to feed it. The Dam remains untouched in respect of scientific investigation which can explore some valuable important information of scientific interests. Phytoplankton is the main micro producers of aquatic environment having photosynthetic pigment chlorophyll. To understand the composition of phytoplankton Different workers have studied the different aspects of phytoplankton and their seasonality in lentic environment . Phytoplankton of lentic water bodies are mainly composed of Cyanophyceae, Chlorophyceae, Bacillariophyceae and Euglenineae.

MATERIALAND METHODS

The sampling of the water of Kanke Dam of Ranchi was done at four selected sites in the month of may and December(2013).Samples were properly collected in plastic bottles following Golterman method and then brought to the laboratory.

BIOTIC FACTORS

Phytoplankton, zooplanktons, Macrophytes - Collection-

Phytoplankton sample were collected in the month of may and December (2013) from three spots of each research site by filtering 85 liters of water with Plankton net of bottling silk number 20 having 75 meshes per linear centimeter. Samples were preserved in plastic bottles in 4% formalin for laboratory studies. Phytoplankton samples of Dam were always collected at the same time on every sampling,to avoid variation in vertical migration during the day.

Counting method-For counting of number of Phytoplankton micro transect method described[12]. 0.1 ml. volume of the sample was taken by using a calibrated medicinal dropper on to a glass slide. A cover slip of known area was placed over it. Slide was observed under microscope and width of high power microscopic filed was measured. Area visible at one focus was considered as one transect, slide was moved from one corner to another, counting the planktons in each microscopic field visible. Several fields were counted by moving the slide in horizontal and vertical directions.

Calculation- No of Plankton/ml = No. of organisms counted in a field x area of cover slip mm2 /Area of one microscopic field x No. of field counted x volume of sample in the cover slip.

For counting individuals pet liter of water method, described by Welch[12] was followed-

$n = a - 100 \times c/L$

where, n = No. of phytoplankton/lit.

a = Average no. of phytoplankton in all count in one cubic millimeter

c = volume of original concentration in millimeter.

L = volume of original water filtered express.

Zooplanktons-

Collection-The zooplankton were collected from different sites, by filtering 5.0 lit of water through a plankton net of bottling silk no.21 with 77 mesh/cm^2 .the zooplankton was preserved in 5% formalin at the site and then brought to the lab for qualitative and quantitative analysis.

Counting- The above preserved sample was taken and made to 10 ml.and thoroughly shaken. One ml,of material was then taken from it and was pieced on a Sedwick Refter Plannkton counting chamber and observed under 40x microscope.

Calculation-The quantitative estimation of zooplankton was made by the following formula given by Welch.

$N= (a \times 1000)c/L$

Where, N= number zooplanktons/liter of water

a = average number of zooplanktons in 1ml of sub. Sample(concentrated)

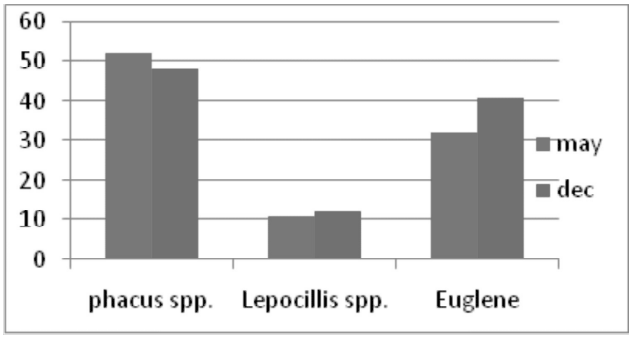
L= volume of original water in liter.

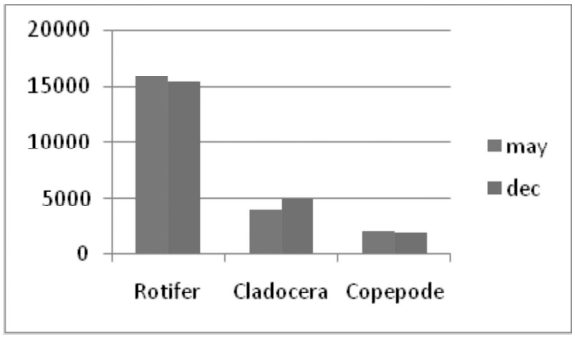
C= ml. of zooplankton concentrate.

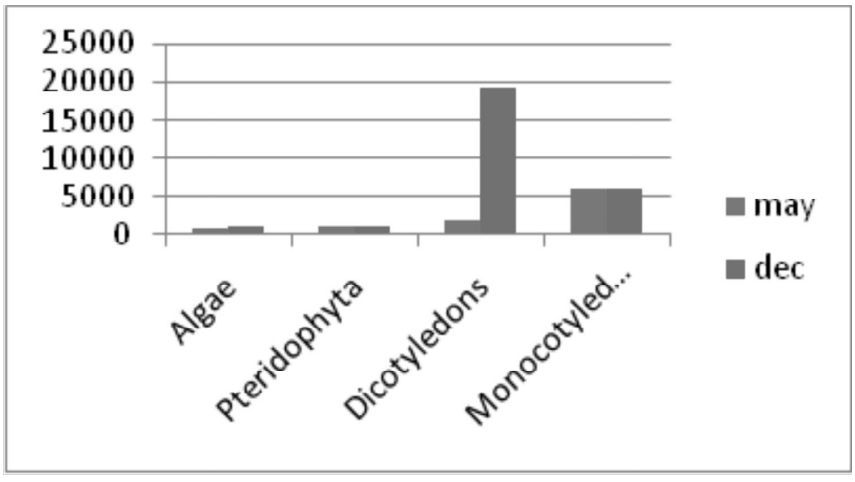
Table1-Annual variation in phytoplankton,zooplankton and macrophytes.

Family	Class	Genus	Variation due to pollution.seasonaly (May & Dec)
Phytoplankton(u/L)	Cyanophyceae Euglenieae	1.Chroococcus. 2.phacus 3.Lepocincilis 4.Euglena	May December(2013) ↑↓ ↓↑
Zooplankton(u/L)	Rotifer Cladocera Copedoda	1.Asplanchna 2. Daphnia	↑↓ ↓↑
Microphytes(u/L)	Algae Prteridophyta Dicotyledons Monocotyledons	1. Chara. 2. Azolla	↑↓ ↓↑

↑ = increases, ↓ =decreases.







Graphs showing variation in (u/L) (A) Phytoplankton,(B) Zooplanktons,(C) Microphytes in the month of may and December (2013)

Identification – the zooplanktons were identified up to species level with the help of monograph and descriptions.

Macrophytes – The microphytes were sampled by using a 0.5 x 0.5 m^2 metel quadrent. Five random qudrent samples taken from each site which were composited and brought to the lab in the polyethylene bags. In the lab all samples were washed with tap water an sorted manually and identified following Gupta.

RESEARCH SITE Kanke Dam of Ranchi of Jharkhand has been taken as the research site for the present study. Kanke Dam was constructed by DVC(1954-1956) Drainage Area 9.44sq mile.Capacity of Reservoir 7.050 ac feet Estimated daily supply 47 Lac gallons Lenght of Dam 4700 feet Maximum Height 53feet The annual rainfall About 1530 mm.

RESULT AND DISCUSSION.

Phytoplanktons are most important biotic factor, it not only indicate the pollution status of water bodies, but also indicate the productivity. But due to pollution and anthropogenic activity variation occurs in phytoplankton, zooplankton as well as macrophytes. In may month Euglenieae shows lowest density and high in December. In zooplankton in may month highest density and lowest in December. Microphytes shows mixed density in both the season.

SUMMARY. In kanke dam a large number of of phytoplankton were recorded, some of them were present throughout in the year while, some where present in particular season. There were altogether 78 genera and 126 species of phytoplanktons water. Population density of phytoplanktons was highest during summer ,lowest during rainy season and modrate during winter season. During the rainy season, the population of zooplanktons was very thin however, no time of the year, there was a total absence of zooplanktons. There were altogether 14 genera and 26 species of zooplancktons. Macrophytes were present in large number in dam. Some were present throught the year, while some were present only in particular season. Macrophytees community comprised of algae, pteridophytes and angiosperms. There were altogether 19 genera and 21 species. Their population densities were highest during late winter to early summer and lowest and during rainy season.

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